TUTORIAL

Umbrella sampling of blocked alanine

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from a run of this tutorial with the 3.2.0 development version.

The first tutorial gave the **potential energy** along the O···H distance of the C7 hydrogen bond of blocked alanine: one minimized structure per point, at 0 K. Here we compute the **free energy** along the same distance at 300 K, the potential of mean force (PMF), with umbrella sampling (US):

1. one molecular dynamics run (a **window**) per structure of the scan, each with a harmonic restraint holding the distance near its value;
2. a check of the windows in the **Umbrella Sampling Browser**: do they overlap? did they stay near their targets?
3. the PMF, joining the windows with WHAM.

You need the system and the scan `bALA_scan_OH.ptGeo` of the first tutorial. The run (20 windows, 70 ps each) takes about 10 s on 12 cores: bALA has only 22 atoms.

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1. Set up the windows

Make the bALA system active and open **Simulate → Umbrella Sampling** (Figure 1).

1. **Input Type**: **From Trajectory (parallel)**, and choose the folder `bALA_scan_OH.ptGeo` in **Input Trajectory**. EasyHybrid reads the reaction coordinate from the `output.log` of the scan and fills **Reaction Coordinate 1** with it (simple distance O5–H17). Each structure of the scan starts one window, whose target is the distance measured in that structure: 20 windows, from 2.06 to 5.85 Å, 0.2 Å apart.
2. **Force Constant**: **100** kJ mol^{-1} Å^{-2}. (Why so weak? See section 2.)
3. **Number of CPUs**: 12.
4. **Molecular Dynamics**: Langevin Dynamics, 300 K, time step 0.001 ps, collision frequency 25 ps^{-1}; **Number of Steps**: **Equilibration** 20000 (20 ps) and **Data Collection** 50000 (50 ps). The **Trajectory Frequency** of the data collection saves the distance at every step (**ptRes** 1) and the coordinates every 100 steps (**ptGeo** 100).
5. **Name** `bALA_US`, **Working Folder** the folder of the system. Click **Run**.

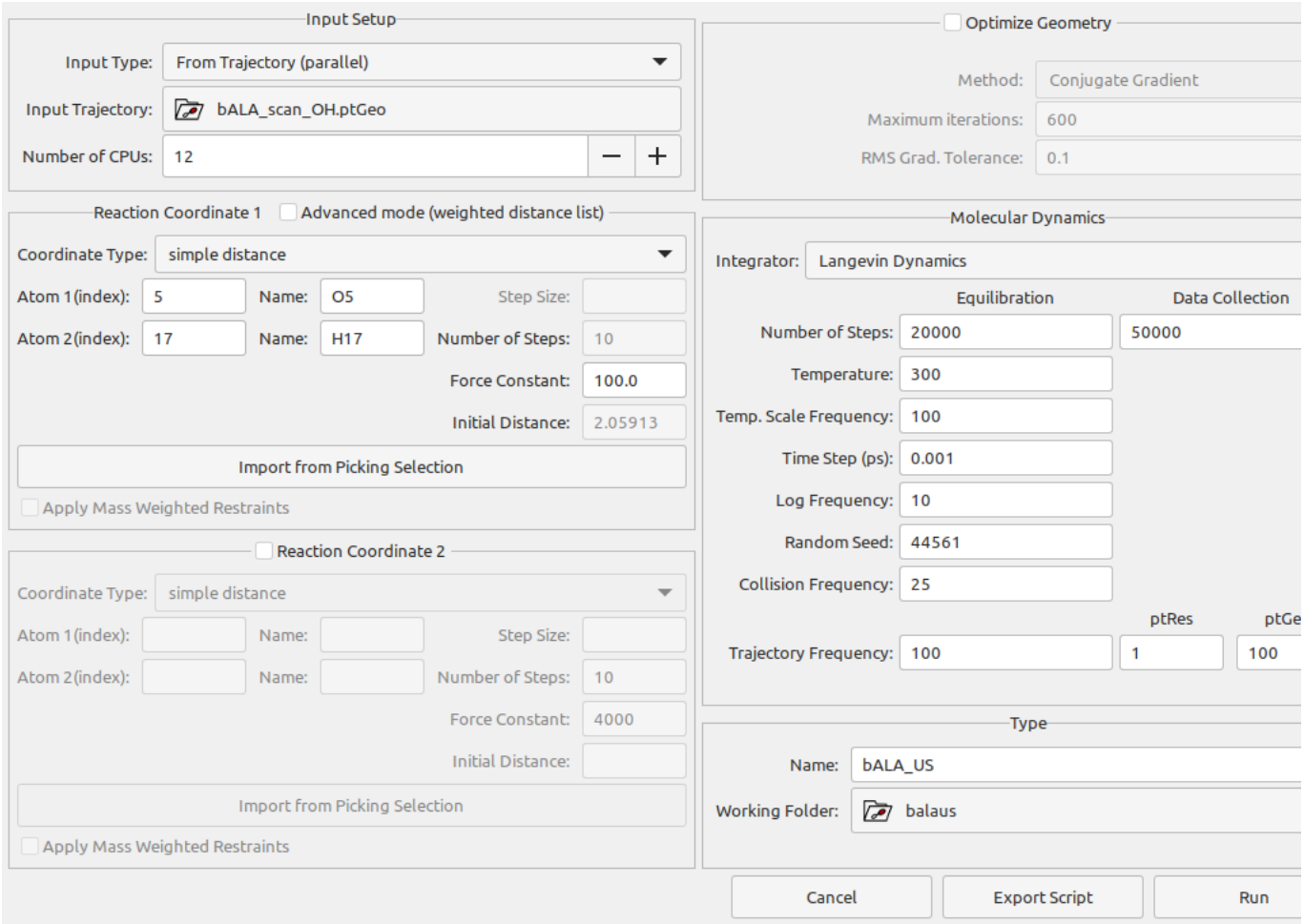


Figure 1. Windows started from the 20 structures of the scan.

The run writes `bALA_US/` with the restraint data of each window (`data_collection/window<i>.ptRes`: the distance at every step) and its coordinates.

2. Force constant and window spacing

The windows must overlap: the distances sampled by each window must have a good part in common with those of its neighbours. On a flat energy surface a harmonic restraint $k(x - x_0)^2$ lets the coordinate spread by $\sigma = \sqrt{(k_BT / 2k)}$:

k (kJ mol^{-1} Å^{-2})	σ at 300 K (Å)	overlap of windows 0.2 Å apart
4000 (a typical scan restraint)	0.018	none
400	0.056	0.07
100	**0.11**	**0.36**

With 0.2 Å between the windows, k = 100 gives an overlap of about 0.4, well above the 0.2 that is usually enough. A stronger restraint would need more windows.

3. Check the windows: the Umbrella Sampling Browser

Analysis → Umbrella Sampling Browser... and choose the `bALA_US` folder. The top of the window summarizes the run (20 windows, 300 K, 50001 restraint samples per window). Look at the diagnostics before any free energy:

- **RC distributions and overlap** (Figure 2): one histogram per window, evenly spaced, each overlapping its neighbours. The table gives the overlap of each pair: 0.36–0.40 here.

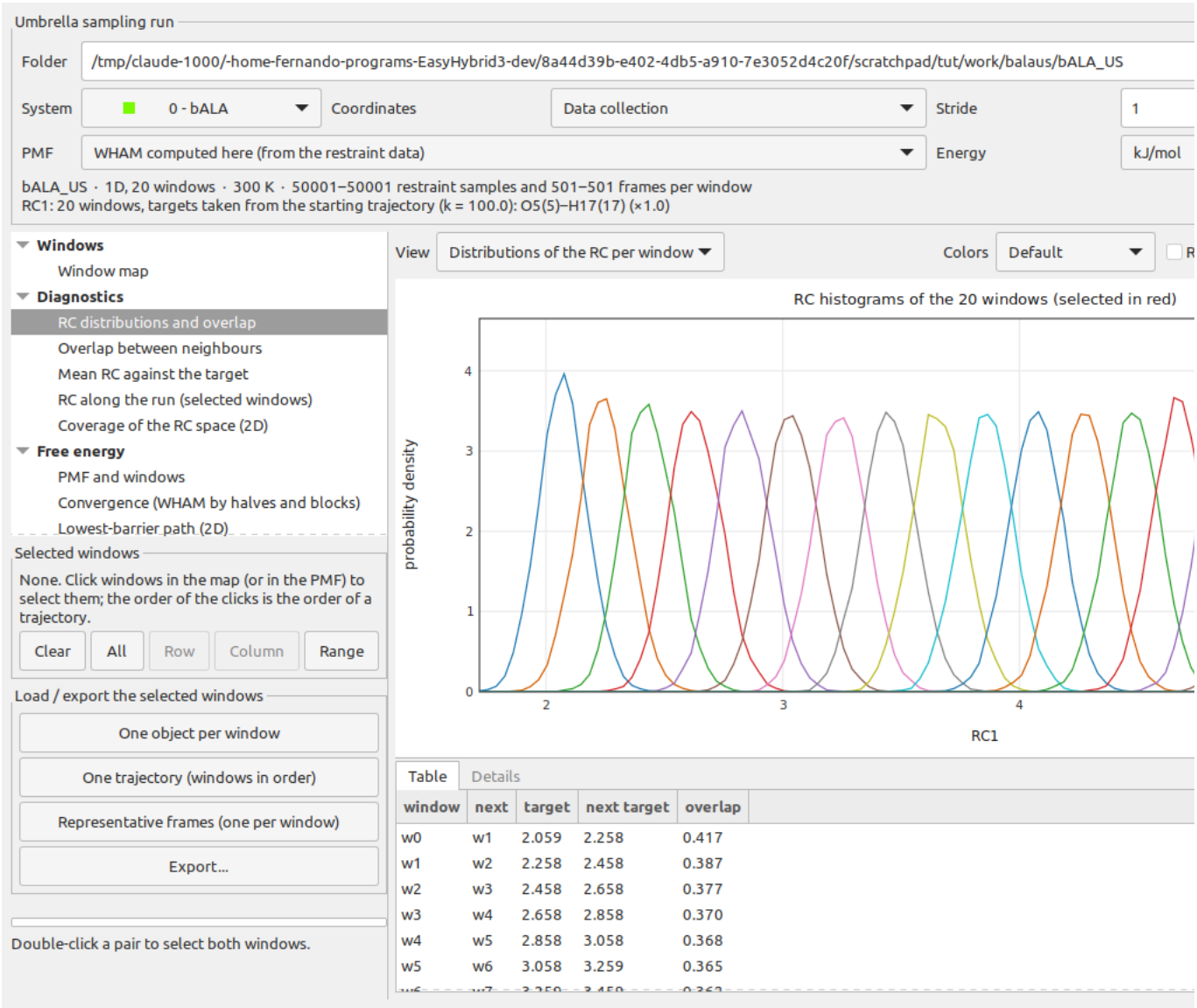


Figure 2. The distance sampled by each of the 20 windows.

- **Overlap between neighbours**: the same numbers as a plot; a pair below 0.1 is a gap in the data.
- **Mean RC against the target** (Figure 3): how far each window drifted from its target. On a slope a weak restraint lets the window slide downhill. Up to 5 Å the windows stay within 0.04 Å of their targets (their spread, σ, is 0.1 Å); above 5 Å, where the energy rises steeply, they sit below their targets, the last one by 0.23 Å. WHAM uses the distances actually sampled, so this costs only resolution at the end of the profile.

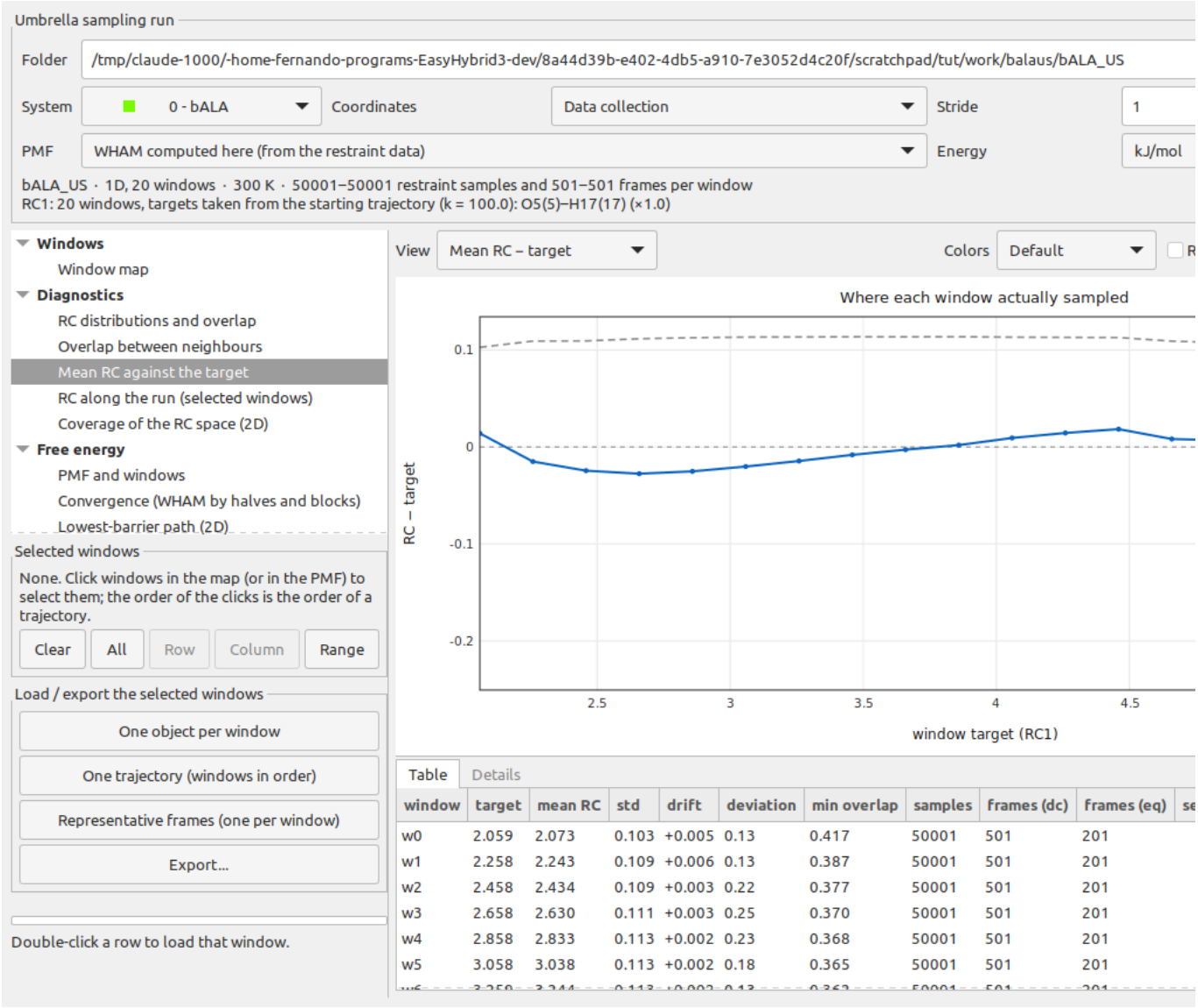


Figure 3. Mean distance of each window against its target.

4. The free energy

PMF and windows (Figure 4) runs WHAM on the restraint data and draws the PMF with the window targets on it.

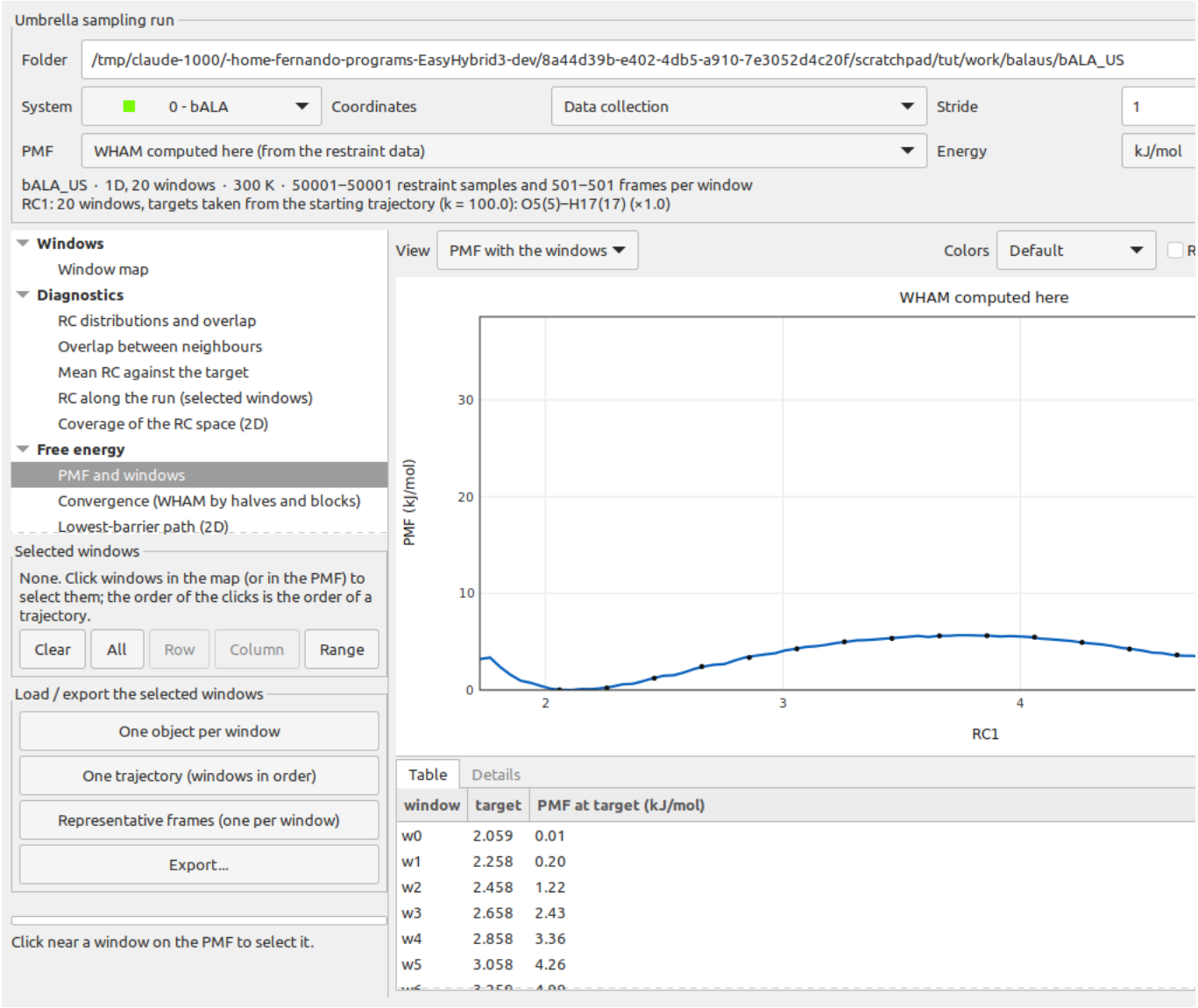


Figure 4. The PMF along the O5···H17 distance at 300 K.

	O···H (Å)	Potential energy (scan)	Free energy (PMF)
C7eq minimum	2.06–2.16	0.0	0.0
Barrier	3.7–3.9	8.3	5.7
C5 minimum	4.9–5.0	5.6	3.1

The free energy follows the potential energy of the scan, about 2.5 kJ/mol (1 kT) lower at the barrier and at the C5 minimum: at 300 K the extended conformers are more flexible (more entropy) than the hydrogen-bonded ring.

The same PMF is given by **Analysis → WHAM** (pDynamo's WHAM): add the `.ptRes` files of `data_collection`, 100 bins, 300 K, and **Run**; it writes `output_pmf.log` and plots the PMF.

Notes

- **Convergence.** *Convergence (WHAM by halves and blocks)* in the Browser compares the PMF of the first and second halves of the data: they should agree within a fraction of a kJ/mol.
- **Windows from a scan** start from relaxed structures that already have the right distance: they equilibrate quickly. With **From Coordinates (sequential)** the windows run one after the other, each starting from the end of the previous one.
- The tutorial QC/MM free energy of the SN2 step in LinB applies the same protocol to an enzyme, where each window costs much more and the choice of k and spacing matters.

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3